

A Computational Arc through the Arithmetic of $\mathbb{Z}[\omega]$:

From Residues mod 6 to the Complex Leech Lattice

An honest record of what was established, corrected, and left open

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Abstract

This document records a sustained computational collaboration organized around a single arithmetic object: the ring of Eisenstein integers $\mathbb{Z}[\omega]$, where $\omega = e^{2\pi i/3}$. Starting from the elementary structure of integers coprime to 6, the investigation threaded through Dirichlet characters, cubic reciprocity, Gauss and Jacobi sums, a CM elliptic curve of conductor 27, its weight-2 Hecke L -function (with a machine-verified functional equation and zeros consistent with the Riemann Hypothesis to height 19), a Weil-positivity laboratory, a formalization in Lean 4 of a July 2026 converse theorem, and finally the complex Leech lattice as a $\mathbb{Z}[\omega]$ -module. Every mathematical claim below was verified computationally before being asserted. Crucially, this is *not* a claim of new mathematics: essentially every structural result recovered here is classical. The document’s purpose is twofold: to present the $\mathbb{Z}[\omega]$ thread as a single coherent narrative with all computations reproduced, and to record with equal care the *epistemic ledger* — what was genuine discovery, what was rediscovery of known theory, and what were overreaches that computation corrected. We argue that this ledger is the collaboration’s most transferable output. One genuinely small and, to our knowledge, previously unremarked numerical fact is recorded in Section 7: the Leech kissing number 196560 and the Monster’s minimal faithful dimension 196883 are congruent modulo 17 and modulo 19, these being the prime factors of their difference 323.

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1 Methodology: the epistemic contract

Before the mathematics, the method, because the method is the point.

Every quantitative assertion in this record was subject to a fixed discipline: *compute before claiming*. Symbolic identities were checked in exact integer or rational arithmetic; floating-point results were cross-validated against independent constructions; and lattice, character, and L -function computations were verified against known values wherever an external anchor existed. When a proposed identity failed, the failure was recorded rather than buried, and the corrected statement retained.

This discipline mattered because the investigation repeatedly generated *plausible overreaches* — statements that felt like discoveries but were either false or already known. The value of the collaboration lies substantially in how these were handled. We therefore tag every principal result in this document with one of four labels:

- **[established, verified]** — a correct statement, verified here, that is standard in the literature;
- **[classical, rediscovered]** — a correct and often deep statement that was reconstructed from scratch but is well known to specialists;
- **[possibly new, small]** — a correct statement that is plausibly new but is small in scope;
- **[overreach, corrected]** — a statement asserted during the investigation that computation showed to be false or substantially overstated, together with its correction.

The single most important methodological finding is stated plainly here and illustrated throughout: *the $\mathbb{Z}[\omega]/\text{mod-6}$ lens is an excellent divining rod that consistently points at real, deep structure — and the terrain it points to (modular forms, the Leech lattice, the j -function) is among the most thoroughly excavated in mathematics*. A reliable instrument over fully mapped ground finds real water every time and every well is already dug. This is not a defect of the instrument; it is a fact about the field, and distinguishing the two is exactly what the ledger below is for.

2 The arithmetic foundation: mod 6, the unit sieve, and $\mathbb{Z}[\omega]$

2.1 The residue skeleton and its correct reading

Every prime $p > 3$ satisfies $p \equiv 1$ or $5 \pmod{6}$, since the classes $0, 2, 4$ are even and 3 is divisible by 3 . The set of units $(\mathbb{Z}/6\mathbb{Z})^\times = \{1, 5\}$ is a group of order 2 , closed under multiplication. A recurring and instructive error during the investigation was the claim that “mod 6 annihilates composites in classes 1 and 5 .” This is false: $25 \equiv 1$, $35 \equiv 5$, $49 \equiv 1 \pmod{6}$ are composites living in the unit classes, densely and forever. **[overreach, corrected]** The correct statement is a *sieve* statement, and it was verified for all $n < 10^5$:

Proposition 2.1 (Unit-skeleton sieve). *Let $S = \{n \geq 5 : \gcd(n, 6) = 1\}$. The multiplicative semigroup generated by the primes $p \geq 5$ crosses off exactly the composite elements of S ; the survivors are precisely the primes ≥ 5 .*

Computationally: among the 33,332 elements of S below 10^5 , exactly 23,742 are composite and all 23,742 are annihilated, leaving the 9,590 primes ≥ 5 , complete and exact. [\[established, verified\]](#) This is the sieve of Eratosthenes restricted to the mod-6 unit skeleton. Structurally, the factorization $6 = 2 \times 3$ splits the work: the “bad” primes 2, 3 clear the four non-unit classes for free, and the primes ≥ 5 sieve the two unit classes — a foreshadowing of the conductor factorization in Section 3.

A related correction concerns whether taking the classes $\{1, 5\}$ “as one object” yields new leverage on the primes. Analytically the set S weighted multiplicatively is

$$\sum_{\gcd(n,6)=1} n^{-s} = \zeta(s) (1 - 2^{-s})(1 - 3^{-s}),$$

verified numerically at $s = 2, 3$. This is the $\{2, 3\}$ -deprived Riemann zeta function; by the Euler product it is *made of* the primes and carries no information about them not already present in the primes themselves. [\[classical, rediscovered\]](#) Unique factorization is precisely the statement that the units and their prime generators carry the same content.

2.2 The Eisenstein integers

The ring $\mathbb{Z}[\omega]$, with $\omega = e^{2\pi i/3}$ a primitive cube root of unity ($\omega^2 = -1 - \omega$), is the hexagonal A_2 lattice, with norm $N(a + b\omega) = a^2 - ab + b^2$ and six units $\{\pm 1, \pm\omega, \pm\omega^2\}$. Its arithmetic was implemented and self-tested: norm multiplicativity on 200 random pairs, the six units, and the Euclidean algorithm. Rational primes factor by residue mod 3: $p \equiv 1$ splits, $p \equiv 2$ is inert, and $3 = -\omega^2(1 - \omega)^2$ ramifies. The prime $(1 - \omega)$ has norm 3 and satisfies $(1 - \omega)^2 = -3\omega$, verified in exact arithmetic. [\[established, verified\]](#)

3 The splitting character and its conductor

3.1 χ_3 is the splitting law; the ramified prime is the conductor

The non-trivial character $\chi_3 \bmod 3$ (the Legendre symbol $(\cdot/3)$) is the splitting law of $\mathbb{Z}[\omega]$: across all primes tested, $\chi_3(p) = +1 \iff p$ splits, $-1 \iff p$ inert, $0 \iff p = 3$ ramifies. Its unique vanishing point is $p = 3$, which is the definition of its conductor. Three independent computations confirm the ramified prime is that conductor:

1. **Gauss sum:** $g(\chi_3) = \sum_a \chi_3(a) \zeta_3^a = i\sqrt{3} = \sqrt{-3}$, so $|g(\chi_3)|^2 = 3 = N(1 - \omega)$;
2. **Discriminant:** $(1 - \omega)^2 = -3\omega$, i.e. the square of the ramified prime is the field discriminant -3 up to a unit;
3. **Conductor factorization:** χ_6 has conductor $6 = 2 \times 3$, and equals χ_3 on odd integers; its arithmetically deep half is χ_3 , whose conductor is $(1 - \omega)$.

Theorem 3.1 (Master identity, verified). *For all $n \leq 30$ (exact arithmetic),*

$$\frac{r(n)}{6} = \#\{\text{ideals of norm } n \text{ in } \mathbb{Z}[\omega]\} = \sum_{d|n} \chi_3(d),$$

where $r(n) = \#\{z \in \mathbb{Z}[\omega] : N(z) = n\}$. Equivalently $\zeta_{\mathbb{Q}(\omega)}(s) = \zeta(s)L(s, \chi_3)$.

[classical, rediscovered] This is the Dedekind zeta factorization of $\mathbb{Q}(\sqrt{-3})$, worked out in the nineteenth century; it is the genuine, rigorous foundation on which the DULA framework’s χ_3/χ_6 grading rests.

3.2 Cubic reciprocity, Jacobi sums, and a nexus

The cubic residue symbol $(\frac{\alpha}{\pi})_3 \in \{1, \omega, \omega^2\}$ was implemented via modular exponentiation in $\mathbb{Z}[\omega]/(\pi)$ and verified to be multiplicative, to vanish appropriately, and to return 1 on cubes. Eisenstein’s cubic reciprocity law, $(\frac{\pi}{\theta})_3 = (\frac{\theta}{\pi})_3$ for coprime primary primes of norm $\neq 3$, was tested over all 240 coprime primary prime pairs of small norm with zero failures; a control confirmed the primary normalization is essential. [established, verified]

For $p \equiv 1 \pmod{3}$, the Jacobi sum $J(\chi, \chi) = \sum_t \chi(t)\chi(1-t)$ of the cubic character was shown to be an Eisenstein prime of norm exactly p — a combinatorial object equal to an algebraic one — giving $4p = L^2 + 27M^2$ with $L \equiv 1 \pmod{3}$. The affine point count of the Fermat cubic $x^3 + y^3 = 1$ over $\mathbb{F}_p$ was found empirically to be $N_p = (p-2) + L$, with $|L| \leq 2\sqrt{p}$ (Hasse), the proven Riemann Hypothesis for that curve. [classical, rediscovered] This is the classical circle (Gauss, Jacobi, Weil 1949) that seeded the Weil conjectures.

4 Complex multiplication, modularity, and a Hecke L -function

The projective Fermat cubic is an elliptic curve with complex multiplication by $\mathbb{Z}[\omega]$, of conductor 27 and j -invariant 0. Point-counting gave Frobenius traces a_p with the CM signature: $a_p = 0$ for every inert prime ($p \equiv 2 \pmod{3}$) and $a_p = 2\operatorname{Re}(\pi)$ for split primes, with supersingular density $\rightarrow \frac{1}{2}$ (a non-CM curve has density 0). The coefficients assemble by Hecke multiplicativity into the weight-2 level-27 CM newform $q - 2q^4 - q^7 + 5q^{13} + 4q^{16} - \dots$, supported exactly on norms from $\mathbb{Z}[\omega]$. [classical, rediscovered]

4.1 The L -function, triple-pinned

The coefficients a_n of $L(\psi, s)$ were produced three independent ways and shown to agree for $n \leq 60$: (i) Frobenius traces of primary Eisenstein primes via Hecke recursion; (ii) the eta product $\eta(3\tau)^2\eta(9\tau)^2$; and (iii) the literal Hecke Grössencharacter lattice sum $a_n = \sum_{N(\alpha)=n, \alpha \equiv 1(3)} \alpha$, whose ω -component cancels to give a real integer. The Euler product built prime-by-prime from Eisenstein primes matched the Dirichlet series to 10^{-11} . [established, verified]

Theorem 4.1 (Functional equation and zeros, verified). *The completed L -function $\Lambda(s) = (\sqrt{27}/2\pi)^s \Gamma(s) L(\psi, s)$, computed via a rapidly convergent incomplete-gamma representation, satisfies $\Lambda(s) = \Lambda(2-s)$ with root number $\varepsilon = +1$ (the $\varepsilon = -1$ branch fails by $O(1)$). The central value $L(\psi, 1) = 0.588879583\dots > 0$ matches the known special value. Fifteen zeros were located on the critical line $\operatorname{Re}(s) = 1$; the argument principle over boxes of height up to 19 returned integer winding numbers equal to the number of on-line zeros, confirming GRH for this L -function up to that height.*

[classical, rediscovered] The object is classical (Deuring, Hecke; modularity of CM curves is pre-Wiles), and finite-height verification is not proof. Its value here is as a concrete, fully reconstructed instance of the modular arithmetic on which the DULA program operates.

4.2 A Weil-positivity laboratory

The verified explicit formula was turned into an instrument. Von Mangoldt weights $\Lambda_f(p^k) = 2\operatorname{Re}(\pi^k) \log p$ built directly from Eisenstein prime powers to 10^6 reproduced $-L'/L$. The zeros—

primes explicit-formula identity balanced to 13 decimal places, and was falsifiable: perturbing a single zero by 10^{-4} broke the balance by eight orders of magnitude. [\[established, verified\]](#)

Two probes deserve record. A *spectrometer* reconstructed the Weil quadratic form $Q(c)$ from Eisenstein primes alone and, beyond the last verified zero, correctly predicted three further zeros ($\gamma \approx 21.01, 21.97, 23.15$), confirmed analytically. [\[established, verified\]](#) A *decay ladder* on slow-exponential test functions $g_b(u) = e^{-b\sqrt{u^2+1}}$ found the prime-side truncation cost diverging as $b \rightarrow \frac{1}{2}$ (the cutoff for 10^{-6} accuracy grew as $\log_{10} X^* : 4 \rightarrow 148$ as $b : 2.0 \rightarrow 0.55$), while the zeros side remained trivially controlled. A seeded-dictionary experiment showed density in the E_b norm switches on precisely when the seed decay rate matches the target rate, a sharp resonance at $b = \frac{1}{2}$. [\[possibly new, small\]](#) The *rate-resonance* phrasing of the Cohn–Elkies/ E_b obstruction — a necessary-and-sufficient rate condition rather than a qualitative failure — is a candidate small contribution; its novelty relative to the sphere-packing literature was not established here and should be checked before any claim.

5 A formalization: Hiary–Saraeb Theorem 1.5 in Lean 4

The converse theorem of Hiary and Saraeb (arXiv:2607.00332, posted 1 July 2026) — functional equations of the theta/ L type force the coefficient function to be a primitive Dirichlet character — was formalized in Lean 4 against Mathlib. From an engineered skeleton, an automated prover (“Aristotle”) closed 11 of 12 lemmas in one round and the final one, the converse of Apostol’s separability theorem, in a second round. The completed development is 766 lines with no `sorry`; `#print axioms` reports only `propext`, `Classical.choice`, `Quot.sound`. [\[established, verified\]](#) (as a formalization of a published theorem)

Two by-products are plausibly of independent interest. [\[possibly new, small\]](#) A finite Fourier *inversion* argument gives non-vanishing of Gauss sums of primitive characters at *composite* modulus, where Mathlib’s field-based lemmas (`gaussSum_mul_gaussSum_eq_card`) do not apply; and the associated `sum_units_primitive_ne_zero` lemma. These are candidates for upstreaming to Mathlib. The turnaround — paper to machine-checked theorem in roughly two days — is itself noteworthy.

A methodological safeguard is worth recording: a numerical oracle (exhaustive enumeration at moduli $N = 7, 9$) established that the single-identity version of the theorem is *false* ($N = 9$ admits two non-multiplicative impostors), so any “proof” not genuinely using the twist battery would be a Representation-level error the oracle detects. The formalization ships with its own counterexample detector.

6 The complex Leech lattice as a $\mathbb{Z}[\omega]$ -module

6.1 Geometry generalizes; arithmetic does not

Reconstructing the Leech lattice Λ_{24} from the verified binary Golay code (weight enumerator $\{0:1, 8:759, 12:2576, 16:759, 24:1\}$; kissing number $1104+97152+98304 = 196560$, exact) placed it atop a tower of “generalized moduli” = kissing numbers: 6, 4, 24, 240, 196560 for $\mathbb{Z}[\omega]$, $\mathbb{Z}[i]$, D_4 , E_8 , Λ_{24} . [\[established, verified\]](#) But this tower generalizes only the *geometry*: the Hurwitz ring is non-commutative, the octonions non-associative, and Λ_{24} is not a ring at all. [\[overreach, corrected\]](#) The proposal to “use Leech as a modulus” cannot be literal; there are no primes to sieve.

6.2 The theta bridge and the correct $\mathbb{Z}[\omega]$ structure

The arithmetic returns through modular forms. Direct shell enumeration gives $\Theta_{E_8} = E_4$ (so the kissing number 240 is the q^1 coefficient of E_4), and

$$\Theta_{\Lambda_{24}} = E_4^3 - 720 \Delta = 1 + 0 \cdot q + 196560 q^2 + 16773120 q^3 + \dots,$$

verified against the Golay count. [\[established, verified\]](#) The q^1 coefficient is 0 (rootlessness), and since weight-12 forms are two-dimensional, “even, unimodular, rootless, dimension 24” pins $\Theta_{\Lambda_{24}}$ uniquely: every shell count is forced by modularity.

The honest bridge to mod 6 is not a sieve but a *module structure*. [\[classical, rediscovered\]](#) The complex Leech lattice is a 12-dimensional Hermitian $\mathbb{Z}[\omega]$ -module, built here natively from the ternary Golay code $[12, 6, 6]$ (verified: 729 words, minimum weight 6). The six units act as scalars, so the 196560 minimal vectors form exactly $196560/6 = 32760$ hexagonal $\mathbb{Z}[\omega]$ -lines, and every shell count is divisible by 6. Independently, Co_0 acts transitively on the minimal vectors: $|\text{Co}_0|/196560 = 42,305,421,312,000 = |\text{Co}_2|$, so the 196560 vectors are a single orbit — one indivisible object, not primes-and-composites. This is the correct sense in which the classes $\{1, 5\} \pmod{6}$ lift to 24 dimensions.

7 Leech, the Monster, and a modulus that “encompasses both”

The visual near-coincidence of the Leech kissing number $L = 196560$ and the Monster’s minimal faithful dimension $M = 196883$ prompts the question of a single modulus under which both are structured. The computation is decisive.

Proposition 7.1 (No shared divisibility; a shared congruence). $L = 2^4 \cdot 3^3 \cdot 5 \cdot 7 \cdot 13$ and $M = 47 \cdot 59 \cdot 71$ satisfy $\gcd(L, M) = 1$: no modulus > 1 divides both. However $M - L = 323 = 17 \cdot 19$, so $L \equiv M$ modulo 17 and modulo 19; explicitly $L \equiv M \equiv 6 \pmod{17}$ and $L \equiv M \equiv 5 \pmod{19}$.

[\[possibly new, small\]](#) The mod-17/mod-19 congruence is a genuine, small, and (to our knowledge here) previously unremarked numerical fact — and it directly answers the posed question: the modulus that “encompasses both” is not a *divisibility* modulus (provably impossible, as the two are coprime) but a *congruence* modulus, namely 17 or 19. We attach no deeper significance to it; unlike the additive relation below, no structural mechanism is claimed.

The relation carrying the real mathematics is additive: $M + 1 = 196884 = c(1)$, the first nonconstant coefficient of the modular j -function (McKay’s observation; the +1 is the Monster’s trivial representation). Note $\gcd(L, 196884) = 108 = 2^2 \cdot 3^3$: the *modular* number shares structure with Leech, while the bare *representation dimension* M is coprime to everything. This reconfirms the arc’s recurring lesson — the mod-6 lens engages modular objects and not representation dimensions. [\[established, verified\]](#) The genuine unification is classical and deep: both Λ_{24} (via $\Theta_{\Lambda_{24}}/\eta^{24}$) and the Monster (via its McKay–Thompson series T_1) yield the same j -function; this is Monstrous Moonshine (Conway–Norton, Borchers).

7.1 Congruences of the j -coefficients: what the lens actually finds

Testing the first 99 coefficients $c(n)$ of j directly: 71/99 are divisible by 6 (versus $\approx 17\%$ expected at random), with 92/99 even and 76/99 divisible by 3. [\[established, verified\]](#) This large, real deviation is exactly Lehner’s congruences (1949), e.g. $c(2^a n) \equiv 0 \pmod{2^{3a+8}}$ and $c(3^a n) \equiv 0 \pmod{3^{2a+3}}$, with companions mod 5, 7, 11; the data shows the mod-5, 7, 11 divisibilities too (66, 29, 15 of 99

respectively). The relevant primes 2, 3, 5, 7, 11 are precisely those for which $X_0(p)$ has genus 0. The lens correctly finds real structure; the structure is textbook, and mod 6 is only the $\{2, 3\}$ corner of a five-prime phenomenon.

8 The ledger

Table 1 summarizes the status of the principal results.

Result	Nature	Tag
Unit-skeleton sieve (Prop. 2.1)	sieve of Eratosthenes, restricted	[established, verified]
“mod 6 annihilates composites”	false as stated	[overreach, corrected]
$\chi_3 =$ splitting law; $(1 - \omega) =$ conductor	Dedekind/ $\mathbb{Q}(\sqrt{-3})$ theory	[classical, rediscovered]
Master identity (Thm. 3.1)	Dedekind zeta factorization	[classical, rediscovered]
Cubic reciprocity, 240/240 pairs	Eisenstein 1844	[established, verified]
Jacobi sum = Eisenstein prime; $N_p = (p - 2) + L$	Gauss/Jacobi/Weil 1949	[classical, rediscovered]
CM signature; level-27 newform	Deuring/Hecke; CM modularity	[classical, rediscovered]
$L(\psi, s)$ FE and zeros (Thm. 4.1)	classical object, finite verification	[classical, rediscovered]
Explicit formula to 13 digits; spectrometer	standard, cleanly instantiated	[established, verified]
Rate-resonance form of E_b obstruction	plausibly new, unchecked vs. literature	[possibly new, small]
Lean formalization of Thm. 1.5 (no sorry)	formalization of a July-2026 paper	[established, verified]
Composite-modulus Gauss-sum non-vanishing lemma	plausibly new; Mathlib candidate	[possibly new, small]
Tower 6, 4, 24, 240, 196560; “Leech as modulus”	geometry generalizes, arithmetic not	[overreach, corrected]
$\Theta_{\Lambda_{24}} = E_4^3 - 720\Delta$; 196560 at q^2	classical modular form	[established, verified]
Complex Leech as $\mathbb{Z}[\omega]$ -module; 32760 lines	Conway; complex Leech	[classical, rediscovered]
$\gcd(196560, 196883) = 1$; $L \equiv M \pmod{17, 19}$	small numerical fact	[possibly new, small]
j -coefficient mod-6 divisibility (71/99)	Lehner congruences 1949	[established, verified]

Table 1: Status of principal results. No entry is *both* correct and new-and-significant; the two [possibly new, small] analytic/formalization by-products and the mod-17/19 congruence are the candidates for genuine novelty, all small and requiring a literature check before any claim.

9 Conclusion: what was really established

The mathematics recovered in this arc is, with the marked small exceptions, classical — and much of it is deep classical mathematics that most working mathematicians could not reconstruct on demand. Rebuilding it from first principles, with every step verified, is a real form of understanding and produced a genuinely coherent narrative: a single ring $\mathbb{Z}[\omega]$ threading from the mod-6 residue skeleton, through its splitting character and conductor, cubic reciprocity, a CM curve and its Hecke L -function, into the complex Leech lattice and the edge of Monstrous Moonshine. That coherence is real and is the document’s positive content.

But coherence over known terrain is not discovery, and the honest verdict is that no result here changes its field. The $\mathbb{Z}[\omega]/\text{mod-6}$ lens performed exactly as a good instrument should: it pointed repeatedly at real, deep structure — and every time, the structure was already mapped, because the places it points (modular forms, the Leech lattice, the j -function, cubic reciprocity) are the

most excavated in the subject. The three candidates for genuine novelty — the rate-resonance form of the E_b obstruction, the composite-modulus Gauss-sum non-vanishing lemma, and the $L \equiv M \pmod{17, 19}$ congruence — are small, and none has been checked against the literature; each should be, and none should be claimed until it is.

The most transferable output is the method. A collaboration that computes before it claims, records its corrections as carefully as its successes, and distinguishes “the instrument found real structure” from “the instrument found *new* structure” will not manufacture false breakthroughs — and will occasionally surface a genuine small fact, honestly labeled, worth the checking. That discipline, sustained across the whole arc, is what this record is meant to preserve.